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Threat Intelligence & Proactive Threat Hunting – AlienVault OTX/OSSIM & YARA

Executive Summary

This module shows how organizations can shift from reactive to proactive security by using cyber threat intelligence (CTI) to detect and hunt threats early. The lab used two free, open-source approaches.

Part 1: OTX Threat Feeds in OSSIM. An AlienVault OTX account was created and its API key was connected to an OSSIM SIEM server (192.168.100.75). OSSIM then downloaded 540 subscribed threat pulses, each containing Indicators of Compromise (IOCs) such as file hashes, IPs, domains and URLs. This allows the SIEM to automatically flag network activity that matches known malicious indicators from the global security community.

Part 2: Threat Hunting with YARA. YARA 4.5.5 was installed on Windows, and both basic and advanced rules were written. The basic rule detected the malicious domain dance4.xyz and the keyword "malware". The advanced rules covered suspicious PE files, malicious document strings, suspicious URLs and IPs (regex), ransomware indicators, and packed files (UPX, aPLib, Themida). Scans matched all rules on the test files. A system scan also produced false positives, such as legitimate Microsoft URLs in procdump64.exe, and a performance warning on the IP regex.

Key Outcomes

- Free community threat intelligence was integrated into a SIEM for early warning.
- YARA rules detected malicious strings, file traits and ransomware indicators.
- Rules must be tested on known-good files to reduce false positives.
- Compiling rules with yarac64 improves performance on large scans.

Integrating OTX Threat Feeds in OSSIM

Learn how to integrate AlienVault Open Threat Exchange (OTX) pulses into an OSSIM (Open-Source Security Information and Event Management) system so you can consume free, community-driven cyber threat intelligence feeds.

Objective

- Create a free AlienVault OTX account and obtain an API key
- Install / access an OSSIM server
- Connect the OTX account to OSSIM
- Download and view subscribed OTX pulses
- Enable automatic detection of network activity matching OTX indicators

Machines Required

Machine	IP Address (example)	Role
Admin / Client	192.168.100.138	Workstation for browser access
OSSIM Server	192.168.100.75	AlienVault OSSIM SIEM

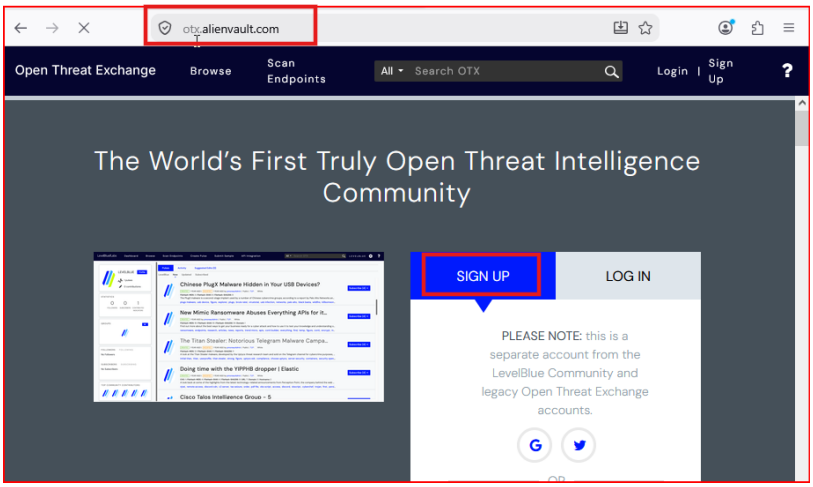
Note: You must install OSSIM yourself. Official ISO downloads are available from AT&T AlienVault / AlienVault community archives. Use a virtual machine (VirtualBox, VMware, Hyper-V, etc.).

Tools Required

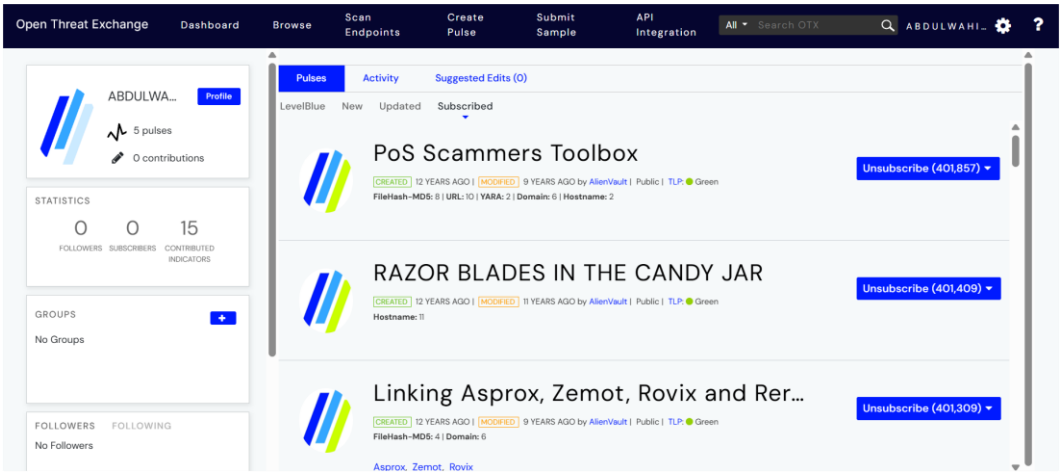
Tool	Purpose
Modern web browser	Access OTX website and OSSIM web interface
AlienVault OTX account	Free threat-intelligence source
AlienVault OSSIM	Open-source SIEM that consumes OTX pulses

Create AlienVault OTX Account & Get API Key

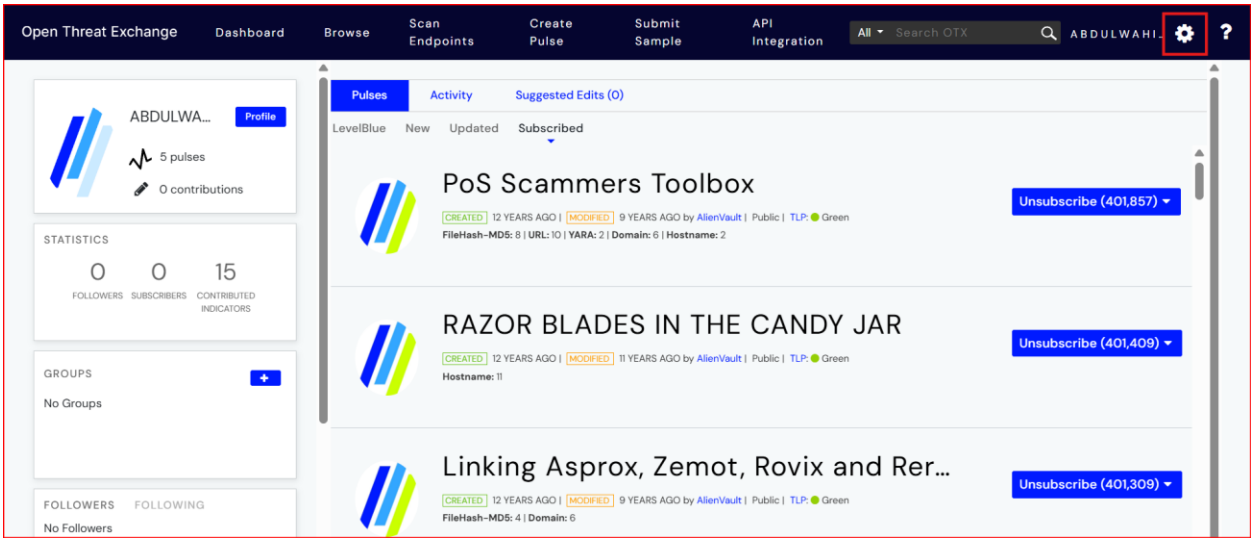
1. On your Admin/Client machine open a browser and go to: https://otx.alienvault.com
2. Click **Sign Up** and complete the registration form with a valid working email address.



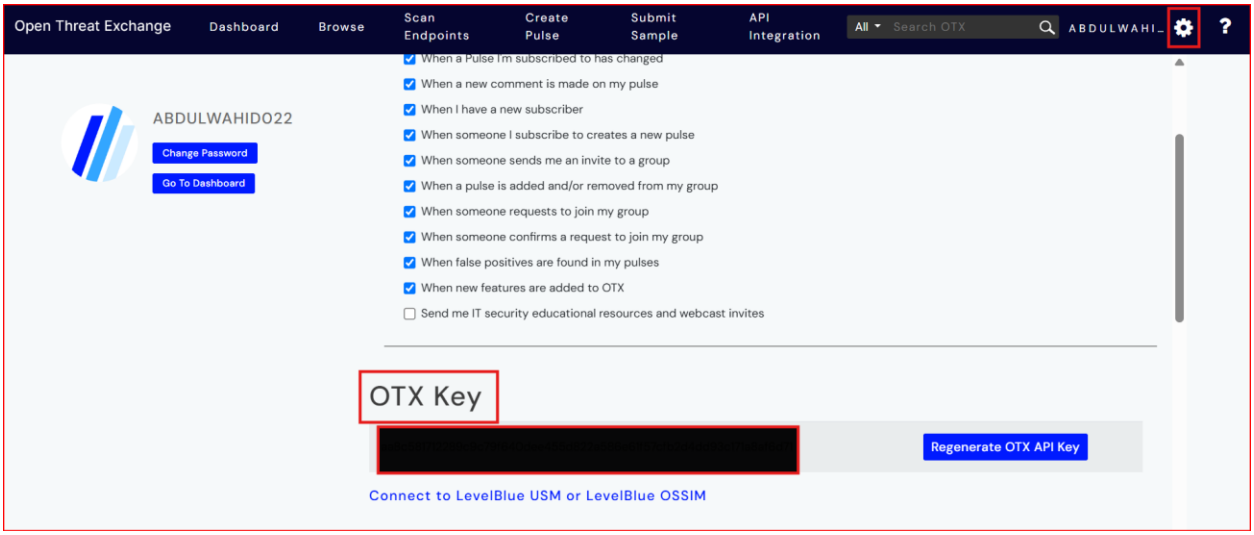
3. Check your email, click the activation link, and activate the account.
4. Log in to OTX with your new credentials.



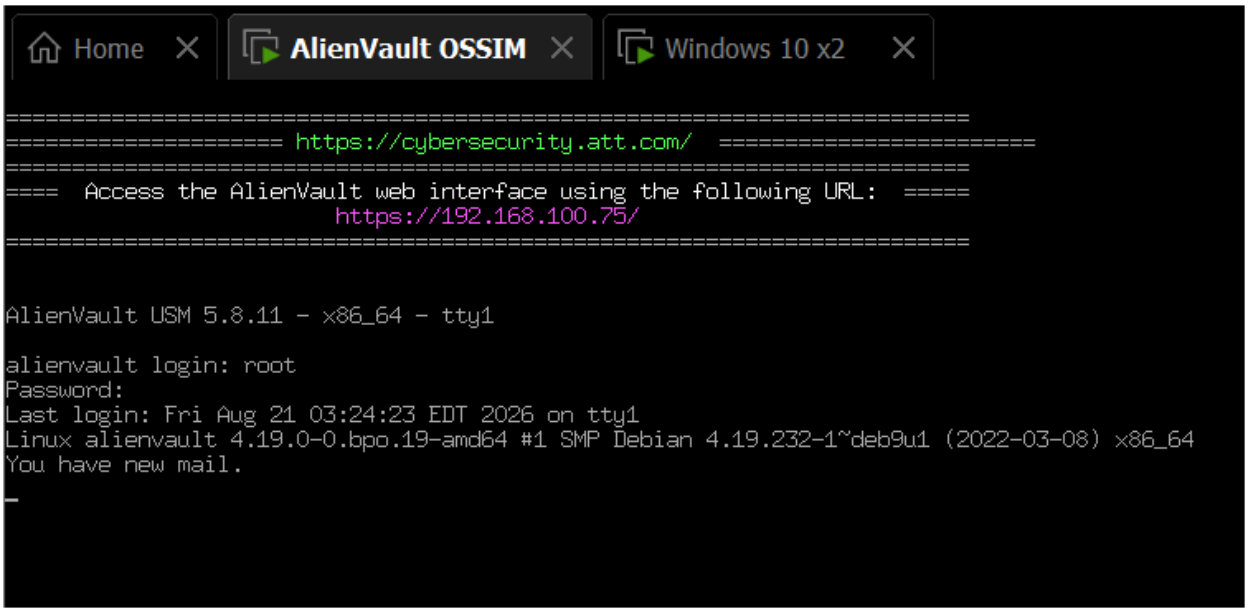
5. Click the profile/settings icon (top-right) → **Settings**.



6. On the Settings page locate the **OTX Key** field and copy the long API key. Keep this key safe – you will paste it into OSSIM later.

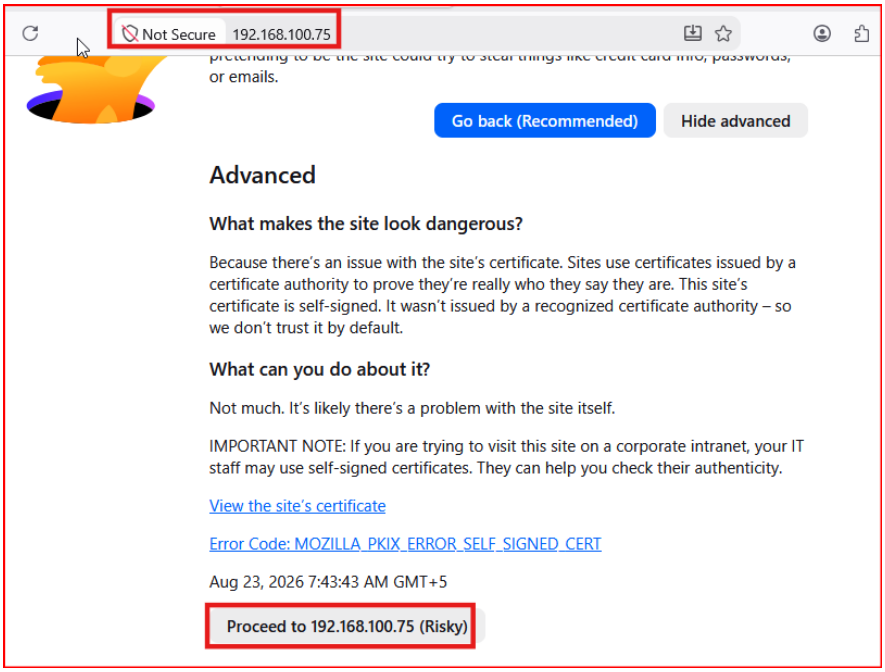


Prepare / Access the OSSIM Server

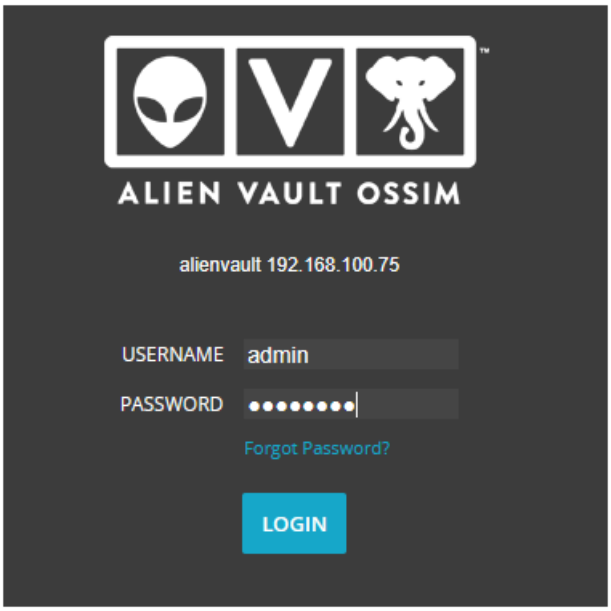


Access the OSSIM Web Interface

1. On the Admin/Client machine open a browser and go to https://<OSSIM-IP> (e.g.https://192.168.100.75)
2. Accept the self-signed certificate warning: Click **Advanced → Proceed to <IP> (unsafe)**.



3. On the login page enter: Username: admin Password: Admin123 Click **LOGIN**.

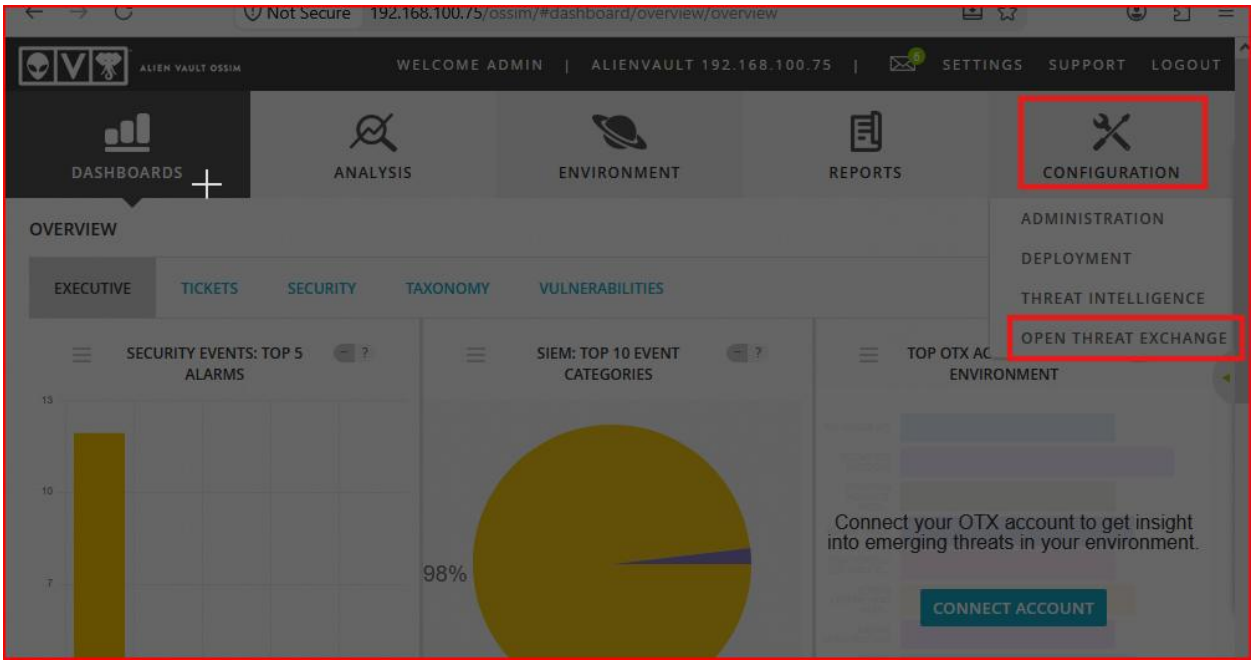


If the "AlienVault OSSIM Getting Started Wizard" appears, click **Skip AlienVault Wizard**.

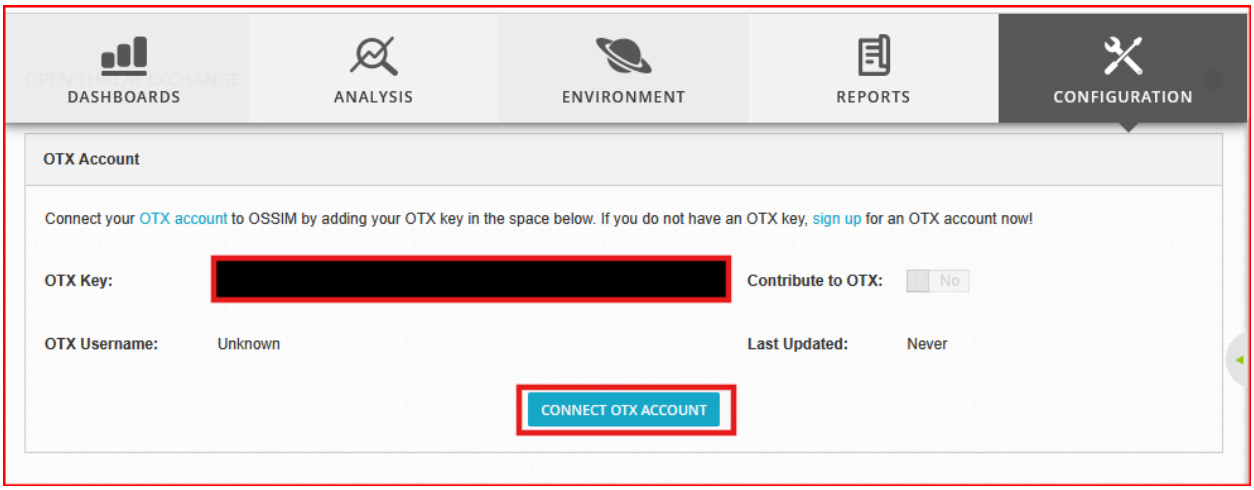
If the "Help us improve AlienVault OSSIM" pop-up appears, click **CANCEL**.

Connect OTX Account to OSSIM

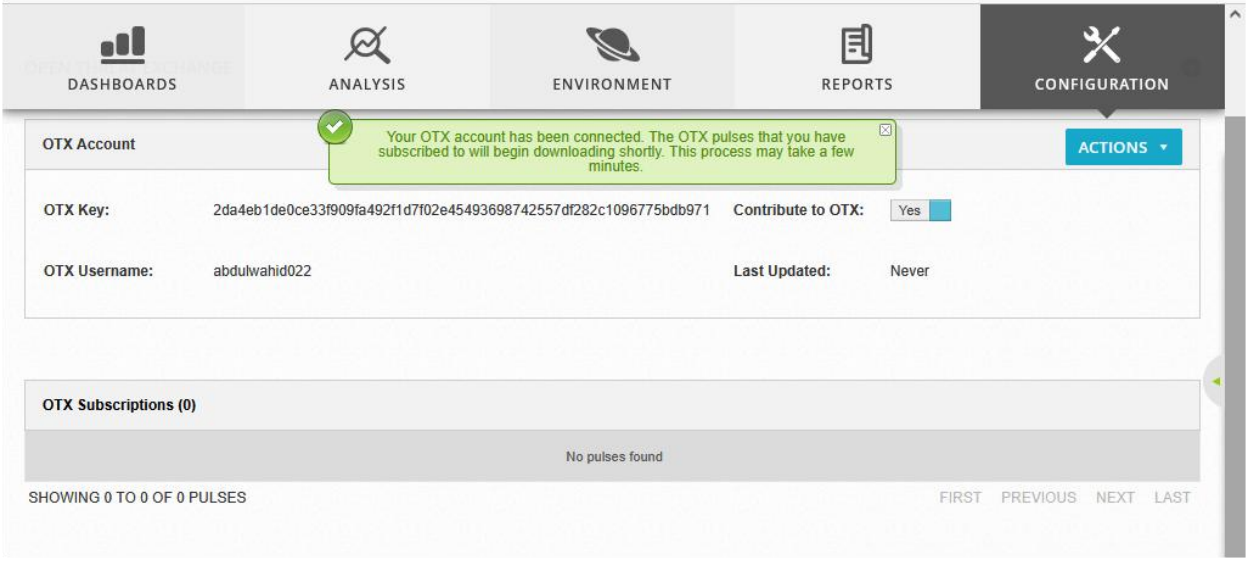
1. In the OSSIM web UI go to the top menu: **CONFIGURATION → OPEN THREAT EXCHANGE**.



2. In the **OTX Key** text box paste the API key you copied earlier.
3. Click **CONNECT OTX ACCOUNT**.

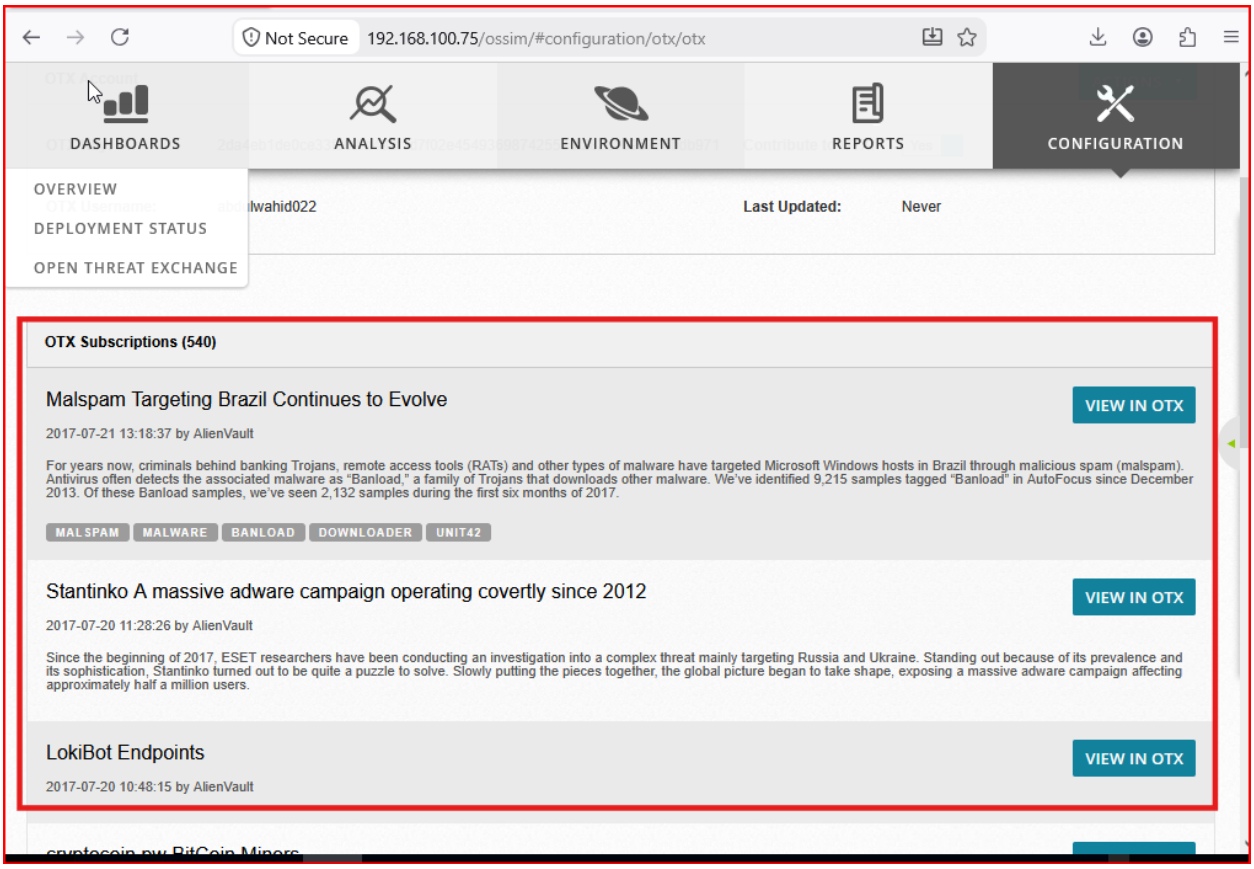


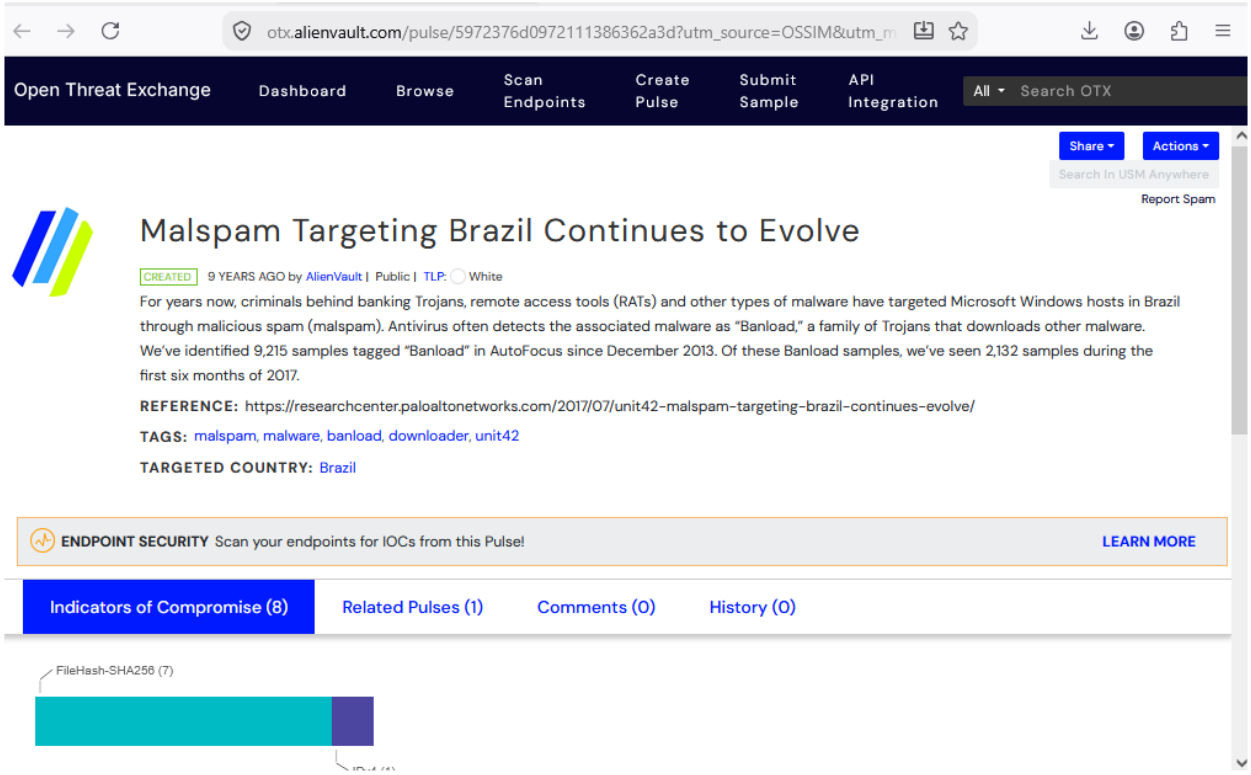
4. After a few seconds the page should show your OTX username and "Contribute to OTX: Yes". The account is now linked.

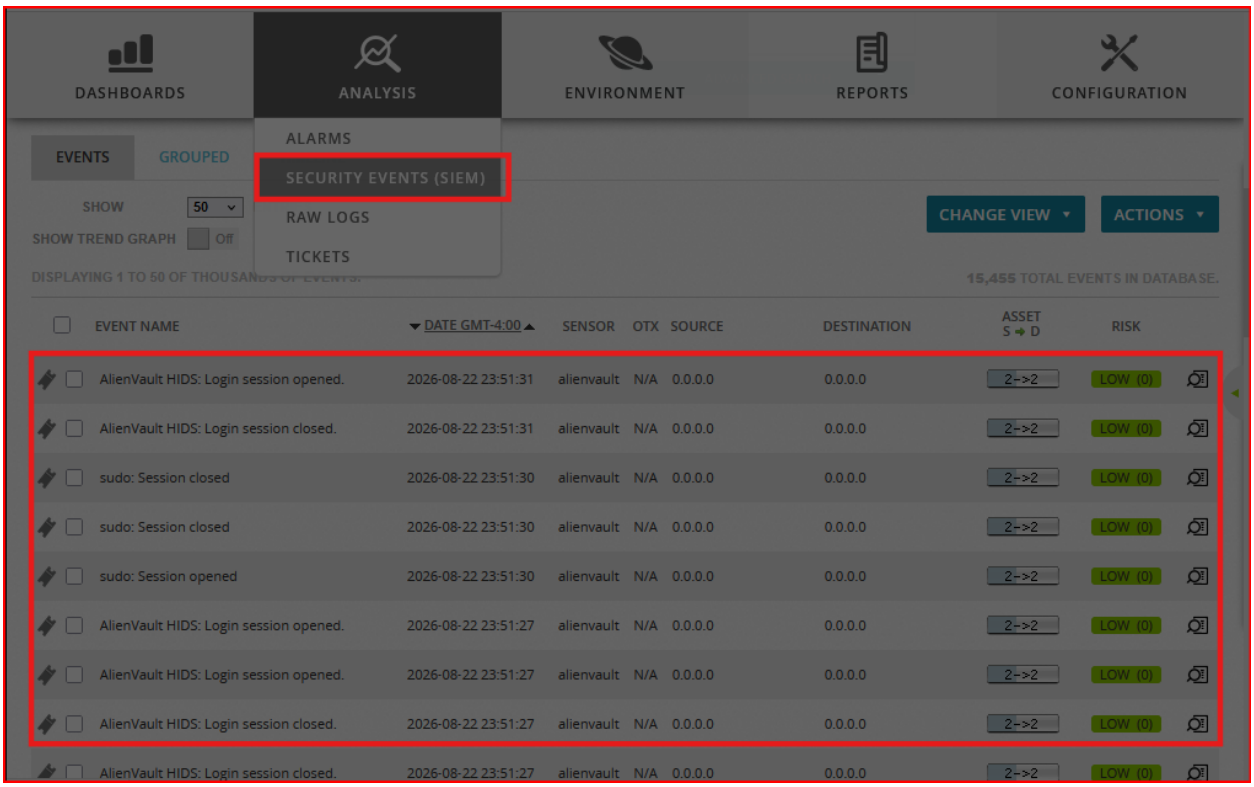


Download & Verify OTX Pulses

1. OSSIM automatically begins downloading the pulses you are subscribed to (or the default set). Download time depends on your internet speed.
2. Under **OTX Subscriptions** you should now see a list of pulses (example: "Real News, Fake Flash – Mac OS X Users Targeted", etc.).
3. You can click **VIEW IN OTX** on any pulse to examine the indicators of compromise (IOCs) on the OTX website.




otx.alienvault.com/pulse/5972376d0972111386362a3d?utm_source=OSSIM&utm_m
Open Threat Exchange
Dashboard
Browse
Scan Endpoints
Create Pulse
Submit Sample
API Integration
All
Search OTX
Share
Actions
Search In USM Anywhere
Report Spam
Malspam Targeting Brazil Continues to Evolve
CREATED 9 YEARS AGO by AlienVault | Public | TLP: White
For years now, criminals behind banking Trojans, remote access tools (RATs) and other types of malware have targeted Microsoft Windows hosts in Brazil through malicious spam (malspam). Antivirus often detects the associated malware as "Banload," a family of Trojans that downloads other malware. We've identified 9,215 samples tagged "Banload" in AutoFocus since December 2013. Of these Banload samples, we've seen 2,132 samples during the first six months of 2017.
REFERENCE: https://researchcenter.paloaltonetworks.com/2017/07/unit42-malspam-targeting-brazil-continues-evolve/
TAGS: malspam, malware, banload, downloader, unit42
TARGETED COUNTRY: Brazil
ENDPOINT SECURITY Scan your endpoints for IOCs from this Pulse!
LEARN MORE
Indicators of Compromise (8)
Related Pulses (1)
Comments (0)
History (0)
FileHash-SHA256 (7)


DASHBOARDS
ANALYSIS
ENVIRONMENT
REPORTS
CONFIGURATION
ALARMS
SECURITY EVENTS (SIEM)
RAW LOGS
TICKETS
EVENTS
GROUPED
SHOW 50
SHOW TREND GRAPH Off
CHANGE VIEW
ACTIONS
15,455 TOTAL EVENTS IN DATABASE.
EVENT NAME
DATE GMT-4:00
SENSOR
OTX
SOURCE
DESTINATION
ASSET S → D
RISK
AlienVault HIDS: Login session opened. 2026-08-22 23:51:31 alienvault N/A 0.0.0.0 0.0.0.0 2->2 LOW (0)
AlienVault HIDS: Login session closed. 2026-08-22 23:51:31 alienvault N/A 0.0.0.0 0.0.0.0 2->2 LOW (0)
sudo: Session closed 2026-08-22 23:51:30 alienvault N/A 0.0.0.0 0.0.0.0 2->2 LOW (0)
sudo: Session closed 2026-08-22 23:51:30 alienvault N/A 0.0.0.0 0.0.0.0 2->2 LOW (0)
sudo: Session opened 2026-08-22 23:51:30 alienvault N/A 0.0.0.0 0.0.0.0 2->2 LOW (0)
AlienVault HIDS: Login session opened. 2026-08-22 23:51:27 alienvault N/A 0.0.0.0 0.0.0.0 2->2 LOW (0)
AlienVault HIDS: Login session opened. 2026-08-22 23:51:27 alienvault N/A 0.0.0.0 0.0.0.0 2->2 LOW (0)
AlienVault HIDS: Login session closed. 2026-08-22 23:51:27 alienvault N/A 0.0.0.0 0.0.0.0 2->2 LOW (0)
AlienVault HIDS: Login session closed. 2026-08-22 23:51:27 alienvault N/A 0.0.0.0 0.0.0.0 2->2 LOW (0)

Successfully integrated free community threat intelligence from AlienVault OTX into OSSIM. Your SIEM can now automatically detect activity matching known malicious indicators (IPs, domains, hashes, URLs, etc.) published by the global security community.

This strengthens your organization's threat-prediction and early-warning capabilities without any commercial cost.

Threat Hunting in Windows using YARA (Enhanced + Advanced)

Learn how to create, refine, and use both basic and advanced YARA rules to detect malicious content, suspicious strings, file characteristics, and indicators of compromise on Windows systems.

Objective

- Install YARA on Windows
- Write a basic YARA rule (original lab rule)
- Write advanced YARA rules (PE detection, multiple strings, conditions, wildcards, etc.)
- Scan local folders and network shares
- Understand real-world threat-hunting usage of YARA

Machines Required

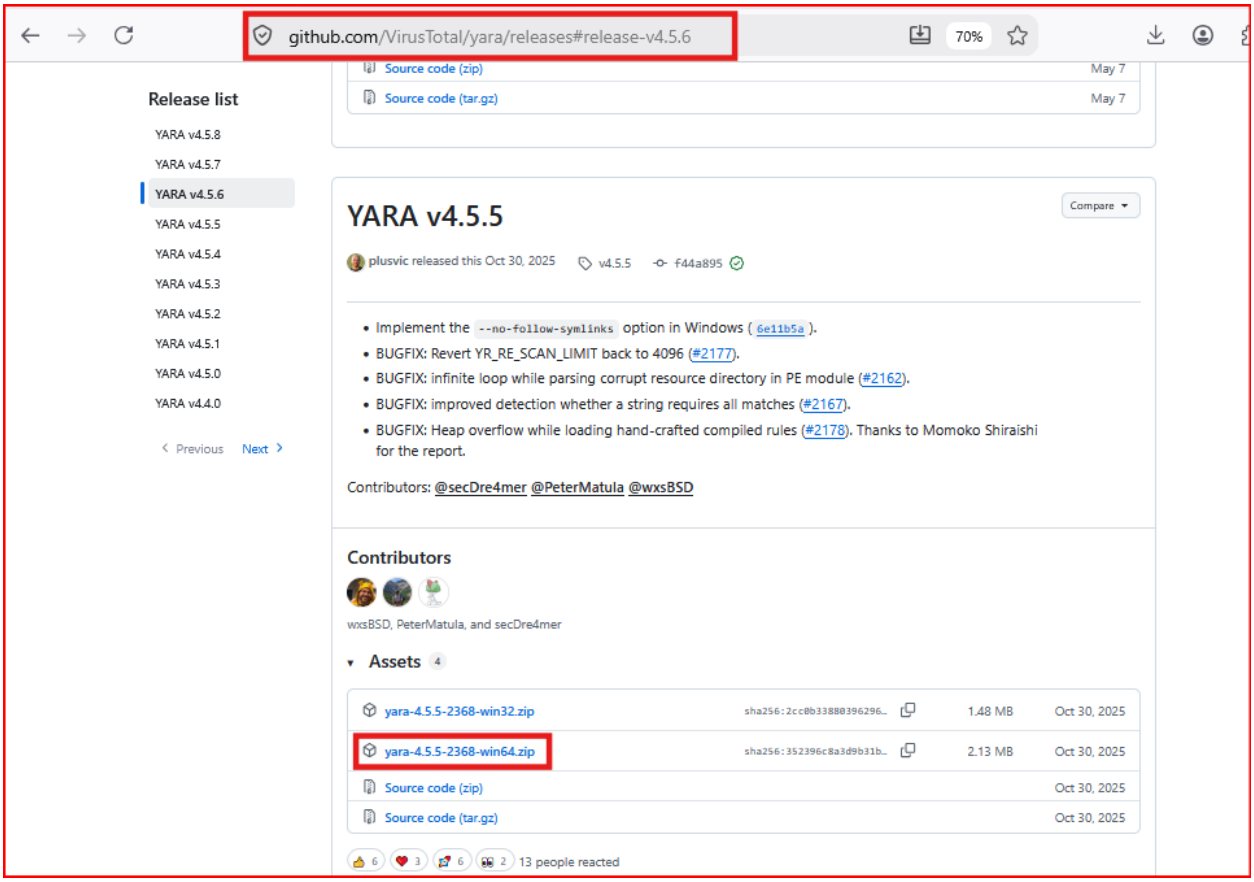
Machine	IP Address (example)	Role
Admin / Client	Your Windows machine	Workstation where YARA runs
Target (optional)	Any Windows share or local folder	Folder containing files to scan

Tools Required

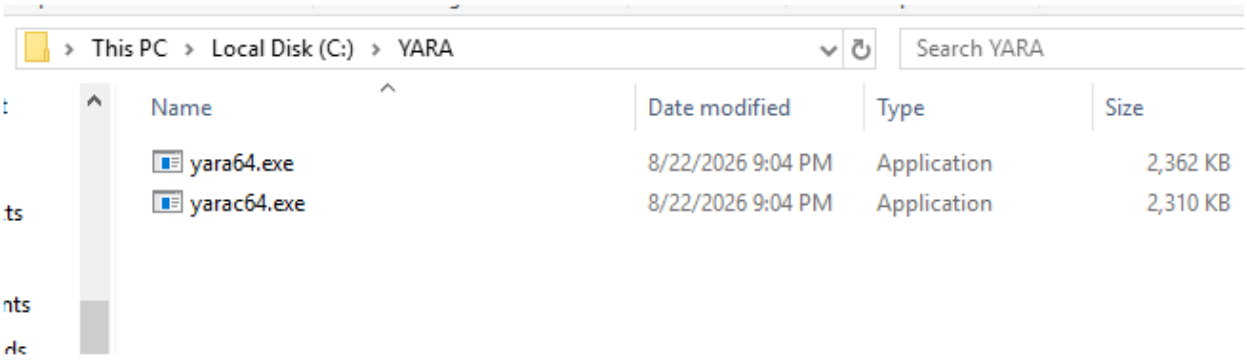
Tool	Purpose	Download / Notes
YARA (Windows 64-bit)	Pattern-matching engine	https://github.com/VirusTotal/yara/releases
Notepad++	Create and edit .yar files	https://notepad-plus-plus.org
Command Prompt	Run scans	Built-in

Download & Install YARA

1. Go to: https://github.com/VirusTotal/yara/releases
2. Download the latest **Windows 64-bit** release (e.g. yara-4.5.x-win64.zip).



3. Create folder: C:\YARA
4. Extract all files into C:\YARA (you should see yara64.exe and yarac64.exe).



Create the Basic Rule

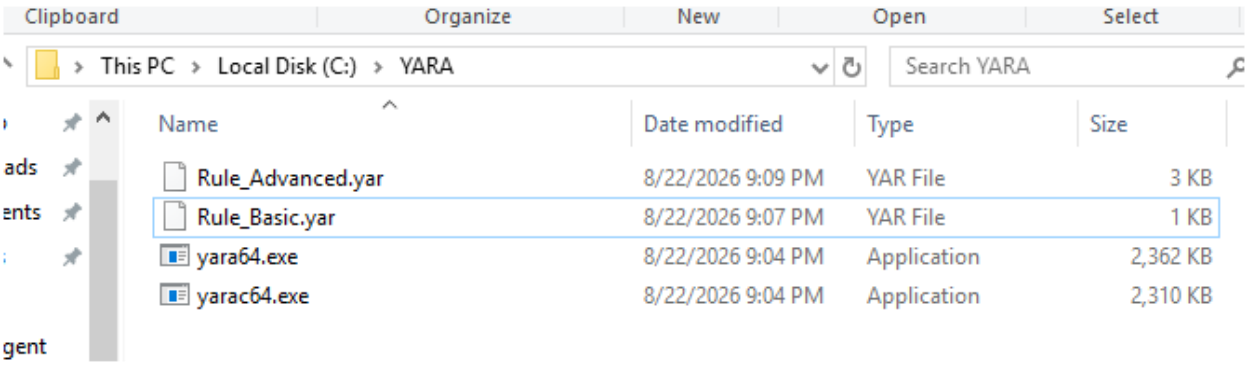
1. Open Notepad++.
2. Paste this rule:

Rule_Basic.txt - Notepad

File Edit Format View Help

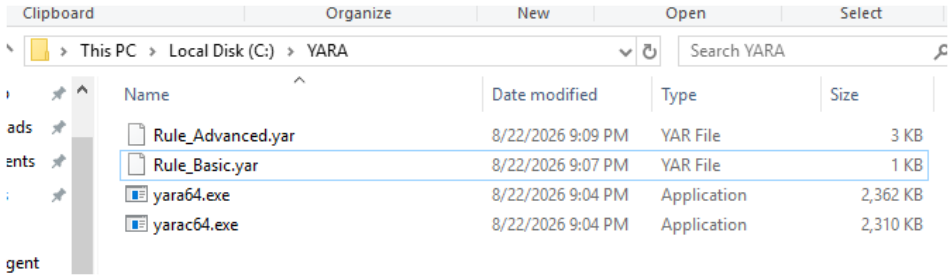
```
rule Malicious_Basic
{
    meta:
        description = "Identify malicious sites - Basic rule from lab"
        author = "abdul"
        date = "2026-11-22"
    strings:
        $str1 = "http://dance4.xyz/" nocase
        $str2 = "malware"
    condition:
        $str1 or $str2
}
```

3. Save as: C:\YARA\Rule_Basic.yar



Create Advanced YARA Rules

Create a new file and paste the following advanced rules. Save as: C:\YARA\Rule_Advanced.yar



Rule_Advanced.txt - Notepad

File Edit Format View Help

```
rule Suspicious_PE_File
{
    meta:
        description = "Detects Windows PE executables that also contain suspicious strings"
        author = "Threat Hunter"
        date = "2026-08-23"
        severity = "High"
    strings:
        $mz = { 4D 5A }                         // MZ header
        $pe = "PE"                              // PE signature
        $sus1 = "cmd.exe" nocase
        $sus2 = "powershell" nocase
        $sus3 = "http://" nocase
        $sus4 = "https://" nocase
        $sus5 = "CreateRemoteThread" nocase
        $sus6 = "VirtualAlloc" nocase
    condition:
        $mz at 0 and $pe and 2 of ($sus*)
}

rule Malicious_Document_Strings
{
    meta:
        description = "Detects documents containing common malware-related strings"
        author = "Threat Hunter"
    strings:
        $s1 = "autoopen" nocase
        $s2 = "document_open" nocase
        $s3 = "shell" nocase
        $s4 = "wscript" nocase
        $s5 = "powershell -enc" nocase
        $s6 = "FromBase64String" nocase
        $s7 = "http://dance4.xyz/" nocase
    condition:
        3 of them
}
```

```
rule Suspicious_URL_and_IP
{
    meta:
        description = "Detects files containing suspicious URLs or IP addresses"
        author = "Threat Hunter"
    strings:
        $url1 = /https?:\/\/[a-zA-Z0-9\.\-]+\/[a-zA-Z0-9\.\-\/_]{5,}/ nocase
        $ip = /\b\d{1,3}\.\d{1,3}\.\d{1,3}\.\d{1,3}\b/
        $bad_domain = "dance4.xyz" nocase
        $bad_domain2 = "malware" nocase
    condition:
        any of them
}
```

```
rule Ransomware_Indicators
{
    meta:
        description = "Common ransomware-related strings"
        author = "Threat Hunter"
        severity = "Critical"
    strings:
        $r1 = "Your files have been encrypted" nocase
        $r2 = "bitcoin" nocase
        $r3 = "decrypt" nocase
        $r4 = "ransom" nocase
        $r5 = ".onion" nocase
        $r6 = "README_FOR_DECRYPT" nocase
    condition:
        3 of them
}
```

```
rule High_Entropy_or_Packed
{
    meta:
        description = "Detects potentially packed or high-entropy files (simple heuristic)"
        author = "Threat Hunter"
    strings:
        $upx = "UPX!"
        $aspack = "aPLib"
        $themida = "Themida"
    condition:
        any of them
}
```

Verify YARA Installation

cd C:\YARA

yara64 --help

```
C:\>cd YARA

C:\YARA>yara64 --help
YARA 4.5.5, the pattern matching swiss army knife.
Usage: yara [OPTION]... [NAMESPACE:]RULES_FILE... FILE | DIR | PID

Mandatory arguments to long options are mandatory for short options too.

      --atom-quality-table=FILE          path to a file with the atom quality table
  -C, --compiled-rules                   load compiled rules
  -c, --count                            print only number of matches
  -E, --strict-escape                    warn on unknown escape sequences
  -d, --define=VAR=VALUE                 define external variable
  -q, --disable-console-logs             disable printing console log messages
      --fail-on-warnings                 fail on warnings
```

Perform Scans

Scan with the Basic rule:

yara64 -s -r Rule_Basic.yar C:\YARA

```
C:\YARA>yara64 -s -r Rule_Basic.yar C:\YARA
Malicious_Basic C:\YARA\Rule_Advanced.yar
0x426:$str1: http://dance4.xyz/
0x2f7:$str2: malware
0x5d4:$str2: malware
Malicious_Basic C:\YARA\Rule_Basic.yar
0xc2:$str1: http://dance4.xyz/
0xef:$str2: malware
```

Scan with Advanced rules:

yara64 -s -r Rule_Advanced.yar C:\YARA

```
C:\YARA>yara64 -s -r Rule_Advanced.yar C:\YARA
warning: rule "Suspicious_URL_and_IP" in Rule_Advanced.yar(45): string "$ip" may slow down scanning
Malicious_Document_Strings C:\YARA\Rule_Advanced.yar
0x350:$s1: autoopen
0x371:$s2: document_open
0x1a5:$s3: shell
0x397:$s3: shell
0x3da:$s3: shell
0x3b5:$s4: wscript
0x3d5:$s5: powershell -enc
0x3fd:$s6: FromBase64String
0x426:$s7: http://dance4.xyz/
Suspicious_URL_and_IP C:\YARA\Rule_Advanced.yar
0x42d:$bad_domain: dance4.xyz
0x5b3:$bad_domain: dance4.xyz
0x2f7:$bad_domain2: malware
0x5df:$bad_domain2: malware
Suspicious_URL_and_IP C:\YARA\yarac64.exe
0x19cb88:$ip: 1.3.6.1
0x19cb8a:$ip: 3.6.1.4
0x19cb8c:$ip: 6.1.4.1
0x19cb8e:$ip: 1.4.1.311
0x19cb90:$ip: 4.1.311.2
0x19cb92:$ip: 1.311.2.1
0x19cb94:$ip: 311.2.1.4
0x19cbe0:$ip: 1.3.6.1
0x19cbe2:$ip: 3.6.1.4
0x19cbe4:$ip: 6.1.4.1
0x19cbe6:$ip: 1.4.1.311
```

```
0x1a7150:$ip: 4.1.311.2
0x1a7152:$ip: 1.311.2.1
0x1a7154:$ip: 311.2.1.12
Suspicious_URL_and_IP C:\YARA\Rule_Basic.yar
0xc9:$bad_domain: dance4.xyz
0xef:$bad_domain2: malware
Ransomware_Indicators C:\YARA\Rule_Advanced.yar
0x6dd:$r1: Your files have been encrypted
0x714:$r2: bitcoin
0x734:$r3: decrypt
0x79d:$r3: DECRYPT
0x61f:$r4: Ransom
0x662:$r4: ransom
0x754:$r4: ransom
0x773:$r5: .onion
0x792:$r6: README_FOR_DECRYPT
High_Entropy_or_Packed C:\YARA\Rule_Advanced.yar
0x89e:$upx: UPX!
0x8b9:$aspack: aPLib
0x8d5:$themida: Themida
```

Perform a scan on your System:

```
C:\YARA>yara64 -s -r Rule_Advanced.yar C:\Users\Public
warning: rule "Suspicious_URL_and_IP" in Rule_Advanced.yar(45): string "$ip" may slow down scanning
Suspicious_URL_and_IP C:\Users\Public\Libraries\RecordedTV.library-ms
0x43:$url1: http://schemas.microsoft.com/windows/2009/library
Suspicious_URL_and_IP C:\Users\Public\procdump64.exe
0x34132:$url1: http://www.microsoft.com/exporting
0x34161:$url1: http://www.microsoft.com/exporting
0x4fb82:$url1: http://crl.microsoft.com/pki/crl/products/MicrosoftTimeStampPCA.crl0X
0x4fbe3:$url1: http://www.microsoft.com/pki/certs/MicrosoftTimeStampPCA.crt0
0x50084:$url1: http://crl.microsoft.com/pki/crl/products/MicCodSigPCA_08-31-2010.crl0Z
0x500e7:$url1: http://www.microsoft.com/pki/certs/MicCodSigPCA_08-31-2010.crt0
0x50550:$url1: http://crl.microsoft.com/pki/crl/products/microsoftrootcert.crl0T
0x505ad:$url1: http://www.microsoft.com/pki/certs/MicrosoftRootCert.crt0
0x50b46:$url1: http://crl.microsoft.com/pki/crl/products/microsoftrootcert.crl0T
0x50ba3:$url1: http://www.microsoft.com/pki/certs/MicrosoftRootCert.crt0
0x51685:$url1: http://www.microsoft.com/pkiops/crl/MicCodSigPCA2011_2011-07-08.crl0a
0x516e6:$url1: http://www.microsoft.com/pkiops/certs/MicCodSigPCA2011_2011-07-08.crt0
0x51d6c:$url1: http://crl.microsoft.com/pki/crl/products/MicRooCerAut2011_2011_03_22.crl0
0x51dd3:$url1: http://www.microsoft.com/pki/certs/MicRooCerAut2011_2011_03_22.crt0
0x51e42:$url1: http://www.microsoft.com/pkiops/docs/primarycps.htm0
0x527ff:$url1: http://crl.microsoft.com/pki/crl/products/MicRooCerAut_2010-06-23.crl0Z
0x52862:$url1: http://www.microsoft.com/pki/certs/MicRooCerAut_2010-06-23.crt0
0x528d0:$url1: http://www.microsoft.com/PKI/docs/CPS/default.htm0
0x52e5d:$url1: http://crl.microsoft.com/pki/crl/products/MicTimStaPCA_2010-07-01.crl0Z
0x52ec0:$url1: http://www.microsoft.com/pki/certs/MicTimStaPCA_2010-07-01.crt0
```

Real-World Usage Tips

- Keep rules updated regularly.
- Combine multiple rules in one .yar file.
- Use yarac64 to compile rules for better performance on large scans.
- Integrate YARA with tools such as Velociraptor, THOR, Loki, or your SIEM.
- Always test new rules on known-good files first to avoid false positives.

Conclusion. Combining OTX and OSSIM for automated, feed-driven detection with YARA for file-level hunting gives a layered, low-cost threat-prediction capability that strengthens early warning and incident response.